so $e$ is an idempotent. To classify other solutions, set $e' = e + c$. For $e'$ to be an idempotent, we must have $ec + ce - c = 0$. This is equivalent to saying that $ece = 0$ and $(1-e)c(1-e) = 0$, so $c = ec(1-e)+(1-e)ce = [e,[e,c]]$. Hence $e' = (1+[c,e])e(1+[c,e])^{-1}$.

Now, in the general case, we prove by induction in $k$ that there exists a lift $e_k$ of $e_{k-1}$ to $A/I^{k+1}$, and it is unique up to conjugation by an element of $1 + I^k$ (this is sufficient as $I$ is nilpotent). Assume it is true for $k = m-1$, and let us prove it for $k = m$. So we have an idempotent $e_{m-1} \in A/I^m$, and we have to lift it to $A/I^{m+1}$. But $(I^m)^2 = 0$ in $A/I^{m+1}$, so we are done. □

**Definition 9.1.2.** A **complete system of orthogonal idempotents** in a unital algebra $B$ is a collection of elements $e_1, \dots, e_n \in B$ such that $e_i e_j = \delta_{ij} e_i$ and $\sum_{i=1}^n e_i = 1$.

**Corollary 9.1.3.** *Let $e_{01}, \dots, e_{0m}$ be a complete system of orthogonal idempotents in $A/I$. Then there exists a complete system of orthogonal idempotents $e_1, \dots, e_m$ ($e_i e_j = \delta_{ij} e_i$, $\sum e_i = 1$) in $A$ which lifts $e_{01}, \dots, e_{0m}$.*

**Proof.** The proof is by induction in $m$. For $m = 2$ this follows from Proposition 9.1.1. For $m > 2$, we lift $e_{01}$ to $e_1$ using Proposition 9.1.1 and then apply the induction assumption to the algebra $(1-e_1)A(1-e_1)$. □

## 9.2. Projective covers

Obviously, every finitely generated projective module over a finite dimensional algebra $A$ is a direct sum of indecomposable projective modules, so to understand finitely generated projective modules over $A$, it suffices to classify indecomposable projective modules.

Let $A$ be a finite dimensional algebra with simple modules $M_1, \dots, M_n$.

**Theorem 9.2.1.** *(i) For each $i = 1, \dots, n$ there exists a unique indecomposable finitely generated projective module $P_i$ such that*

$$\dim \operatorname{Hom}(P_i, M_j) = \delta_{ij}.$$